



Project Proposal

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	25
Understanding	3					
Analysis	4					
Implementation	8					
Report Writing	10					
Total obtained mark						
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Title: YELO (Your Everyday Local Options)

1. INTRODUCTION

1.1 Background Overview

The rental market in Dhaka, particularly in emerging hubs like Uttara, is characterized by high demand from students, young professionals, and freelancers. Currently, this sector operates largely through informal channels such as Facebook groups, WhatsApp communities, and physical brokers. While platforms like Bikroy.com and Bproperty exist, they are primarily designed for buying/selling large real estate assets or general classifieds. They lack specific features tailored for quick, trustworthy rentals of smaller units like bachelor rooms, seats, or office desks.

In the context of Uttara, which hosts major universities and corporate offices, finding a verified space is time-consuming and risky. Existing digital solutions suffer from three main gaps:

1. **Lack of Identity Verification:** Anyone can post fake listings anonymously.
2. **Inefficient Search:** Users must manually filter dozens of parameters without natural language support.
3. **Absence of Intelligence:** No system analyzes price fairness, detects scams automatically, or verifies image quality before publication.

Artificial Intelligence (AI) is currently underutilized in these local platforms. Most sites rely on human moderation, which is slow and costly. There is a significant opportunity to integrate lightweight, zero-cost AI models directly into the web application to automate trust-building and discovery processes.

1.2 Problem Statement

The core problem addressed by YELO is the lack of a secure, transparent, and intelligent platform for renting residential and small-scale commercial spaces in localized areas like Uttara. Specifically:

- **Trust Deficit & Scams:** Users frequently encounter fake listings, phantom landlords, and price manipulation because there is no robust verification mechanism. Current platforms allow anonymous posting without identity checks, leading to financial loss and safety risks.

- **Inefficient Discovery:** Searching for specific amenities (e.g., "generator backup," "metro proximity") requires manual filtering through hundreds of irrelevant posts. Natural language queries in mixed Bengali/English are not supported, forcing users to spend excessive time refining filters.
- **Lack of Accountability:** When disputes arise (e.g., landlord demanding higher rent after contact, or tenant ghosting), there is no lightweight resolution mechanism. This leads to frustration and discourages honest usage.
- **Manual Verification Burden:** Traditional methods require SMS OTPs (costly and slow) or manual admin reviews (scalability issue). There is a gap in using zero-cost, biometric/location-based AI verification.

Why AI Is Needed

Today, identifying fake images, detecting scam keywords, suggesting fair prices, and parsing natural language searches are done manually by users or expensive third-party services. YELO integrates on-device and server-side AI models to automate these tasks. Without AI, the system would remain a static directory prone to spam and inefficiency. With AI, it becomes an active agent that protects users and enhances experience.

1.3 Objectives

General Objective

To develop YELO, a web-based marketplace that securely connects service providers (landlords/office owners) with customers (tenants/freelancers) for renting spaces in Uttara, ensuring transparency, trust, and efficiency.

AI Objective

To implement five core AI components that replace manual effort:

1. Natural Language Processing (NLP): To enable smart search using mixed Bengali/English queries.
2. Computer Vision: To detect low-quality, blurry, or duplicate images at upload time.
3. Anomaly Detection: To identify potential scam listings based on price outliers and suspicious keywords.
4. Statistical Modeling: To provide dynamic price suggestions based on historical sector data.

5. Generative Templating: To auto-generate professional property descriptions from structured inputs.

1.4 Scope

In-Scope Features

- Hyperlocal Focus: Restricted to Uttara Sectors 1–18. This ensures data density and relevance for the MVP.
- User Roles: Renter (Customer), Provider (Landlord/Office Owner), and Minimal Admin (Technical Maintenance only).
- Listing Types: Residential (Flat, Room, Sublet, Seat) and Commercial (Office Desk, Co-working Space).
- Verification System: Location tracking + Live Selfie Capture (No SMS OTP).
- AI Modules: Smart Search, Image Quality Check, Scam Detection, Price Suggestion, Description Generator.
- Dispute Resolution: Community-driven reporting with automated Trust Score penalties.
- Freemium Model: Mock payment integration for "Featured Listings."

Out-of-Scope Features

- Full Payment Gateway: Actual rent transactions will not be processed; YELO is a connection platform.
- Nationwide Coverage: Expansion to other districts is future work.
- Mobile Application: The initial release is a responsive Web App.
- Complex Arbitration: Legal adjudication of disputes is out of scope; the system only flags and suspends accounts.
- Heavy ML Training: We use pre-trained/lightweight models rather than training deep neural networks from scratch.

1.5 Stakeholders

- End Users (Renters): Students, young professionals, and freelancers looking for affordable, verified spaces in Uttara. Their interest lies in speed, safety, accurate pricing, and ease of contact.
- Service Providers: Landlords, co-working space operators, and office managers. Their interest is in generating genuine leads quickly, reducing scam reports, and having easy tools to create attractive listings.
- System Administrator: The development team/maintainers. Responsible for technical health, handling extreme abuse cases (banning users), and monitoring

system logs. Their interest is in maintaining platform integrity with minimal manual overhead.

1.6 Proposed Solution

YELO proposes a decentralized, AI-enhanced marketplace. Unlike traditional classifieds, YELO embeds intelligence into the user journey:

1. Onboarding: Users register via email/password. During registration/login, the browser captures their GPS location and prompts for a live selfie. This creates an immutable "Identity Anchor" without costing money on SMS gateways.
2. Listing Creation: Providers input basic details. The AI Description Generator converts checkboxes (AC, WiFi) into persuasive text. The Image Analyzer rejects blurry/duplicate photos instantly. The Scam Detector flags unrealistic prices before publishing.
3. Discovery: Renters use the Smart Search Bar to type queries like "২ বেডরুমের ফ্ল্যাট সেক্টর ৭ এ ওয়াইফাই সহ". The NLP engine parses this and applies filters automatically. Results can be viewed in List or Interactive Map View.
4. Connection & Safety: To view a provider's contact, the renter must agree to the listed price (creating a Transaction Record). If a dispute arises, either party can file a report. The system automatically reduces the offender's Trust Score. Low scores lead to visibility reduction or suspension.
5. Monetization: Providers can pay a nominal mock fee to make their listing "Featured," pushing it to the top of search results.

1.7 AI Features

Existing AI Limitations in Competitors:

Current platforms (Bikroy, Facebook Groups) rely on human moderation. They do not have image quality checks, meaning users waste time on bad photos. They lack price analytics, leading to information asymmetry. They have no natural language understanding for mixed-language queries.

Proposed AI Features in YELO:

1. Smart NLP Search:
 - Input: Mixed Bengali/English text query (e.g., "cheap room near metro").
 - Output: Structured JSON filter object {sector: 7, type: room, max_price: 5000}.

- Replaces: Manually selecting 5+ dropdown filters.
 - Technique: Keyword Extraction & Regex Parsing (Client-side JS + Server-side validation).
2. Image Quality & Duplicate Check:
- Input: Uploaded image file.
 - Output: Pass/Fail status + Warning message (Blurry/Dark/Duplicate).
 - Replaces: Admin reviewing every photo for clarity/authenticity.
 - Technique: TensorFlow.js (Mobilenet/Laplacian Variance) + Perceptual Hashing.
3. Scam/Spam Detection:
- Input: Listing Title, Description, Price.
 - Output: Risk Score (0-100) + Flag Status.
 - Replaces: Human moderators reading every post for red flags.
 - Technique: Rule-based Engine + TF-IDF Keyword Matching + Statistical Outlier Detection (Python/scikit-learn).
4. Price Suggestion Engine:
- Input: Property Attributes (Sector, Beds, Amenities).
 - Output: Recommended Price Range & Median Value.
 - Replaces: Providers guessing prices, leading to long vacancy periods.
 - Technique: MongoDB Aggregation Pipelines (Historical Data Analysis) + Pandas (Python).
5. Description Generator:
- Input: Checkbox selections (AC, Lift, Parking).
 - Output: Professional Paragraph Text.
 - Replaces: Writing marketing copy for each listing.
 - Technique: Template-Based Dynamic Text Generation (String Interpolation).

Why AI Is Essential

Without these AI features, YELO becomes just another static classified site. The Image Checker prevents spam; the Scam Detector builds trust; the NLP Search improves UX significantly. Removing AI makes the platform unusable for its target audience who demand speed and safety. Thus, AI is the core of the product, not an add-on.

2. SYSTEM REQUIREMENTS

2.1 Functional Requirements

Module 1: User Management & Security

- FR-01: The system shall allow users to register with Name, Email, Phone, and Password.
 - Stakeholder: Renter, Service Provider.
 - Detail: Standard form validation. Password must be hashed using bcrypt before storage.
- FR-02: The system shall capture the user's current geographical location during registration and store it for security purposes.
 - Stakeholder: Renter, Service Provider.
 - Detail: Uses Browser Geolocation API. Requires explicit user consent. Stored as Latitude/Longitude pair.
- FR-03: The system shall prompt the user to take a live selfie via webcam/mobile camera during registration.
 - Stakeholder: Renter, Service Provider.
 - Detail: Opens camera interface. Captures single frame. Uploads to Cloudinary. Used for identity anchoring.
- FR-04: The system shall assign a "Verified" badge to users who successfully complete location and selfie capture.
 - Stakeholder: Renter, Service Provider.
 - Detail: A visual indicator appears on the profile upon successful completion of FR-02 and FR-03.
- FR-05: The system shall calculate a dynamic Trust Score (0-100) based on verification status, review ratings, and dispute history.
 - Stakeholder: System Administrator (Logic definition), All Users (Visibility).
 - Detail: Initial score 70. +10 for Verified Badge. Adjusted by +/- points based on Reviews and Disputes.
- FR-06: The system shall log all login attempts with IP address, timestamp, and current geolocation for security auditing.
 - Stakeholder: System Administrator.
 - Detail: Creates a record for security analysis. Allows detection of suspicious activity (e.g., login from different city).

Module 2: Listing Management (Provider Side)

- FR-07: The system shall allow providers to create listings categorized as Residential or Commercial.
 - Stakeholder: Service Provider.
 - Detail: Selection of category determines available sub-types (Flat, Room, Desk, etc.).
- FR-08: The system shall restrict location selection to Uttara Sectors 1–18.
 - Stakeholder: Service Provider.
 - Detail: Dropdown menu limited to predefined sectors. Ensures hyperlocal focus.
- FR-09: The system shall enforce a minimum of 5 photos per listing.
 - Stakeholder: Service Provider.
 - Detail: Form submission blocked if fewer than 5 images are uploaded.
- FR-10: Upon photo upload, the system shall run an AI Image Quality Check; if the image is blurry or a duplicate, the upload shall be rejected with a specific error message.
 - Stakeholder: Service Provider.
 - Detail: Client-side TensorFlow.js checks Laplacian variance for blur. Server-side hash check for duplicates. Immediate feedback loop.
- FR-11: The system shall provide an "Auto-Generate Description" button that converts selected amenities into a formatted paragraph.
 - Stakeholder: Service Provider.
 - Detail: Takes checkbox inputs (WiFi, AC, Lift) and injects them into pre-written templates to create professional text.
- FR-12: Before saving, the system shall run an AI Scam Detection algorithm. If the risk score exceeds 60, the listing shall be flagged for admin review or blocked.
 - Stakeholder: Service Provider, System Administrator.
 - Detail: Python backend analyzes text for scam keywords and compares price against sector average. High-risk listings enter "Pending Review" state.
- FR-13: The system shall display a Price Suggestion tooltip when the provider enters a price, showing the average rent for similar properties in that sector.
 - Stakeholder: Service Provider.
 - Detail: Queries database for median price of same Sector/Type/Beds. Displays range (e.g., "Typical: 15k-18k").

Module 3: Search & Discovery (Customer Side)

- FR-14: The system shall provide a global search bar accepting natural language queries in Bengali and English.
 - Stakeholder: Renter.
 - Detail: Input field supports Unicode. Sends string to backend NLP parser.
- FR-15: The system shall parse search queries to extract entities (Location, Type, Price, Amenities) and apply them as database filters.
 - Stakeholder: Renter.
 - Detail: Backend uses regex/dictionary matching to convert "Sector 7 flat" into {sector: 7, type: 'flat'}. Returns filtered results.
- FR-16: The system shall display search results in two modes: Grid/List View and Interactive Map View (using Leaflet.js/OpenStreetMap).
 - Stakeholder: Renter.
 - Detail: Toggle switch changes UI component. Map view plots markers with price tags based on lat/lng.
- FR-17: The system shall allow filtering by "Verified Providers Only."
 - Stakeholder: Renter.
 - Detail: Checkbox filter excludes listings where provider is "Verified Providers Only"
- FR-18: The system shall sort results by Relevance, Price (Low-High), and Featured Status.
 - Stakeholder: Renter.
 - Detail: Default sort is Relevance (Match score). Featured items always appear first if enabled.

Module 4: Connection & Dispute Resolution

- FR-19: The system shall hide provider contact details until the customer clicks "View Contact."
 - Stakeholder: Renter, Service Provider.
 - Detail: Phone number masked (XXX-XXXX-XXX) in public view. Unmasked only via authenticated request.
- FR-20: Clicking "View Contact" shall trigger a modal requiring the customer to check a box agreeing to the listed price.
 - Stakeholder: Renter.
 - Detail: Explicit consent step. Prevents claims of misunderstanding later.
- FR-21: The system shall create an immutable Transaction Record linking the Customer ID, Provider ID, Listing ID, and Agreed Price upon contact reveal.

- Stakeholder: System Administrator (Audit), Renter, Service Provider.
 - Detail: Saves entry in transaction history. Timestamped. Cannot be edited/deleted by users. Basis for dispute resolution.
- FR-22: The system shall allow both parties to file a Dispute Report against a Transaction Record, uploading evidence (screenshots).
 - Stakeholder: Renter, Service Provider.
 - Detail: Form accessible from transaction history. Accepts image uploads for proof.
- FR-23: Upon filing a dispute, the system shall automatically deduct 10 points from the reported user's Trust Score.
 - Stakeholder: System Administrator (Policy enforcement).
 - Detail: Automated penalty. Encourages honest behavior.
- FR-24: If a user accumulates 3+ active disputes, the system shall temporarily suspend their account pending admin review.
 - Stakeholder: System Administrator.
 - Detail: Account status changed to "Suspended". Login disabled. Admin notified.

Module 5: Admin Dashboard (Minimal)

- FR-25: The system shall allow admins to view a list of suspended/banned users.
 - Stakeholder: System Administrator.
 - Detail: Table view of users with status Active/Inactive. Option to Restore or Permanent Ban.
- FR-26: The system shall allow admins to permanently delete fraudulent listings.
 - Stakeholder: System Administrator.
 - Detail: Soft delete option for flagged listings. Removes from public search.
- FR-27: The system shall display basic statistics: Total Users, Total Active Listings, Total Disputes Resolved.
 - Stakeholder: System Administrator.
 - Detail: Simple counters on dashboard home page. No complex analytics charts required.

2.2 Non-Functional Requirements

- NFR-01 (Performance): The system shall load search results within 2 seconds for up to 1,00 concurrent listings.
 - Reason: Critical for user retention. Achieved via efficient indexing and rendering.

- NFR-02 (Security): All passwords shall be hashed securely. API endpoints shall be protected via token authentication.
 - Reason: Prevents credential theft. Ensures only authorized users can access private data (contacts, disputes).
- NFR-03 (Usability): The interface shall be fully responsive, supporting mobile, tablet, and desktop views.
 - Reason: Target audience (students/professionals) primarily uses smartphones. Layout adapts to screen size.
- NFR-04 (Reliability): The AI Image processing shall occur client-side to reduce server load, ensuring functionality even during minor network latency.
 - Reason: Offloads computation to user device. Make system feel faster. Reduces hosting costs.
- NFR-05 (Privacy): User location and selfie data shall be encrypted at rest and never shared publicly. Users can request deletion of their biometric data.
 - Reason: Compliance with ethical standards. Builds trust. Data minimized to essential fields.
- NFR-06 (Cost): The solution shall utilize free-tier cloud services to ensure zero operational cost for the academic prototype.
 - Reason: Feasibility constraint for student project. Must demonstrate sustainability without funding.

3. TOOLS AND TECHNOLOGIES

Programming Languages:

Frontend: JavaScript (ES6+), HTML5, CSS3.

Backend & AI: Python 3.9+.

Frameworks and Libraries:

Frontend Framework: React.js (with Vite for fast bundling).

Styling: Tailwind CSS (for rapid, industry-standard UI design).

Mapping: Leaflet.js + OpenStreetMap (Free, open-source alternative to Google Maps).

State Management: Context API / Redux Toolkit.

Backend Framework: FastAPI (High-performance Python web framework).

ASGI Server: Uvicorn.

AI Models, APIs, and Libraries:

Computer Vision (Client-side): TensorFlow.js (tensorflow-models for classification, custom Canvas API logic for blur detection).

Machine Learning (Server-side):

- scikit-learn: For clustering similar listings and basic anomaly detection.
- pandas & numpy: For statistical analysis (Price Suggestions).
- NLTK or spaCy: For Bengali/English text processing (NLP Search & Spam Detection).

File Storage: Cloudinary SDK (Python package).

Database:

Primary DB: MongoDB Atlas (NoSQL, flexible schema for varied listing attributes).

Driver: Motor (Async driver for MongoDB with Python).

Development & Deployment Tools:

Version Control: Git & GitHub.

Deployment: Vercel (Frontend), Render (Backend).

Testing: Jest (Unit Testing), Postman (API Testing).

4. PROJECT TIMELINE AND WORK PLAN

The project is estimated to take 10 weeks. Tasks are divided among 2 team members. Member A focuses on Frontend & Client-Side AI. Member B focuses on Backend, Server-Side AI, and Database.

Phase	Duration	Key Activities	Deliverables	Assigned To
1. Planning & Design	Week 1	Requirement gathering, SRS writing, UML Diagrams (Use Case, ERD), UI Wireframing.	Finalized SRS Document, Necessary Diagrams.	Both

2. Core Backend Setup	Week 2	Setting up FastAPI, connecting to MongoDB, implementing JWT Auth, creating API endpoints for Listing CRUD.	Working API Endpoints.	Member B
3. Frontend Foundation	Week 3-4	React setup, Routing, Global State, Basic UI Components (Navbar, Footer, Cards).	Responsive Shell Application.	Member A
4. AI Module Integration	Week 5-6	Implementing Scam Detection (using scikit-learn rules), Price Suggestion (using pandas aggregation), and NLP parsing logic in Python backend. Connecting React frontend to these Python APIs.	Functional AI Features integrated with Core.	Member B
5. Search & Map Features	Week 7	Integrating Leaflet Maps, Advanced Filtering UI, Wishlist functionality. Client-side Image Quality Check implementation.	Interactive Map View, Filtered Search Results, Image Validator.	Member A

6. Dispute & Trust System	Week 8	Building Transaction Records, Dispute Reporting Forms, Trust Score Algorithm, Admin Panel basics.	Complete Security Workflow.	Member B
7. Testing & Debugging	Week 9	Unit Tests, Integration Tests, Bug Fixing, Performance Optimization.	Test Reports, Stable Build.	Both
8. Documentation & Demo	Week 10	Final Report Compilation, Presentation Slides, Video Demo Recording.	Final PDF Report, Demo Video, Source Code Repo	Both

5. OPTIONAL SECTIONS

5.1 Existing System

Currently, the primary systems used for rental searches in Bangladesh include:

- Facebook Groups: Highly active but chaotic. No structure, no verification, high scam risk. Posts disappear quickly.
- Bikroy.com / Bproperty: Structured databases but focused on sales or large estates. Poor UX for small rentals (rooms/seats). No AI assistance for search or fraud detection.
- Co-working Websites (Moar, Oasis): Individual websites for specific brands. No aggregator platform exists for comparing desk rents across different providers.

These existing systems fail to provide a unified, secure, and intelligent experience. They lack the ability to verify user identity cheaply, analyze listing authenticity automatically, or understand natural language queries. YELO addresses these gaps by combining the structural benefits of a database-backed site with the intelligence of AI and the accessibility of social media-like verification.

5.2 System Architecture

The system follows a decoupled Client-Server architecture optimized for AI integration.

- **Presentation Layer (Frontend):** Built with React.js. It handles user interaction, displays data, and performs lightweight AI computations locally. Specifically, TensorFlow.js runs in the browser to check image quality (blur/brightness) before sending files to the server, reducing bandwidth and server load. Leaflet.js renders maps using OpenStreetMap tiles.
- **Application Layer (Backend):** Built with FastAPI (Python). It serves RESTful APIs. It contains the heavier AI logic:
 - **NLP Parser:** Receives raw text queries, cleans them, extracts entities (keywords), and returns structured filter objects.
 - **Scam Detector:** Analyzes incoming listing data against historical patterns and keyword blacklists using scikit-learn/pandas.
 - **Price Engine:** Executes MongoDB aggregation pipelines to calculate medians and averages for suggestions.
- **Data Layer:** MongoDB Atlas stores unstructured data for listings (flexible schemas for varying amenities) and structured data for users/transactions. Cloudinary stores media files (images/selfies) securely.
- **Communication:** Secure HTTPS communication between Frontend and Backend. JWT tokens manage session state.

This architecture ensures scalability. As user base grows, the frontend scales independently (CDN), while the backend can handle increased AI computation loads efficiently due to Python's ecosystem strengths.

5.3 Ethical, Legal, and Social Considerations

- **Data Privacy:** Collecting selfies and locations raises privacy concerns. YELO adheres to strict data minimization principles. Biometric data (selfies) is stored separately from personal profiles and is deleted immediately upon account closure. Explicit consent is obtained via pop-ups before accessing camera/GPS. Data is encrypted at rest.
- **Bias in AI:** The Scam Detection algorithm relies on keyword lists and statistical outliers. To prevent bias against certain dialects or phrasing, the keyword database

will be regularly updated and reviewed. The Price Suggestion engine uses historical data; care is taken to exclude anomalous/extreme values to avoid skewing recommendations.

- Fairness: The Trust Score system is transparent. Users can see why their score dropped (e.g., "1 Dispute Filed"). This ensures accountability without black-box algorithms. Appeals process is handled by the Admin, providing a human oversight layer for edge cases.
- Legal Compliance: The platform acts as an intermediary, not a landlord. Terms of Service clearly state that YELO does not guarantee the condition of the property or the conduct of the parties beyond the verification steps provided. Liability limitations are included in the user agreement.

5.4 Future Work

- Expansion: Scaling from Uttara to Bashundhara, Mirpur, and eventually nationwide. The modular database schema allows adding new "Areas" without code changes.
- Advanced AI: Integrating Azure Face API or AWS Rekognition for 1:1 face matching between registration selfie and login selfie for stronger liveness detection. Adding Sentiment Analysis for reviews to gauge overall satisfaction trends.
- Mobile App: Developing native Android/iOS apps using React Native for better camera/GPS access and push notifications.
- Payment Gateway: Integrating SSLCommerz/bKash for booking deposits, moving from a pure connection model to a transactional model.
- Virtual Tours: Allowing 360-degree video uploads for immersive property viewing.

AI Closure: *AI is used for grammatical corrections. The market analysis and strategic insights represent in case of comparison with our original work and deep understanding of the industry and technological stack.*